To: Rocky Mountain Space Grant Consortium

Re: Support for **James Hyland** UNSGC Fellowship

From: Todd K. Moon, Electrical and Computer Engineering Department

Date: June 5, 2023

I am happy to support James Hyland's application for the Space Grant Fellowship. James has been working with me on Doppler-based target tracking, and has made significant contributions to this project.

In this project, an incident radio wave is used as a source for a bistatic tracking system. We are prototyping this in an indoor setting, and using WiFi signals as the radio source. But this can extend to a much broader scenario. For example, the principles can be adapted to planet wide geolocation using radio signals emitted from orbiting satellites. In fact, early geolocation systems used these methods. This may be useful as planetary/lunar exploration moves ahead. Satellites placed around Mars or the Moon may serve the double function of communication satellites and navigation satellites, building upon principles developed in this research.

As a result of this research, we hope to develop journal publications (to extend beyond conference publications already produced) and significantly move James toward completion of his masters degree in the following year.

Match may come from a variety of sources. There are research funds that I will be drawing on personal research funds in the Electrical and Computer Engineering department. I will also be exploring departmental graduate student support, as well as contributing my own time toward this research project.